Experimental Determination of Sump Pump Performance Curves

Group A2

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Final Report

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ME – 360 Engineering Experimentation

Abstract

A sump pump (an Everbilt 1/6 HP Plastic Utility Pump) was tested by measuring the flow rate of the pump, the pressure from the system, and the power consumed by the pump. These measurements were taken across various tests, across a range of voltages from 95V to 125V, each set of tests providing raw data that was then used to calculate the pump head and efficiency. These were analyzed against flow rate to determine a line of best fit. The best efficiency point (BEP) for each voltage case was calculated for the pump, yielding a range of calculated efficiencies from 12% to 19%, with pressure head ranging from 0.9m to 8m. It was determined that operating the pump at 100V 60Hz AC at a flow rate of 38.47 ± 0.62 LPM and a head value of 5.288 ± 0.471m offered the optimal balance of flow rate and efficiency, determined by a comparative analysis of the specific energy consumption, total dynamic head, and efficiency at the BEP.

Introduction

Context

Sump pumps are pumps utilized to remove water that collects in “sump basins or sump pits” (SUMP & SEWAGE PUMP MANUFACTURERS ASSOCIATION, 2013). Generally, sump basins or pits are spaces designed such that water can collect; particularly In the residential or industrial context, sumps are specifically allocated for and designed to collect excess water in properties such that it can be pumped elsewhere. While the usage may vary depending on the application, such as in crawlspaces in the foundation of homes, basement floors, or in industrial cooling towers, nonetheless the usage of sump pumps allows for proper operation of sumps for protection against water damage. There are different ways in which pumps are constructed, the most common of which is the centrifugal pump. These pumps have a rotating impeller to push fluid typically directly connected to an electrical motor.

As with any other mechanical device, pumps have variable efficiencies dependent on the load applied to the device. These are characterized in a pump performance curve (map), which describes the relationship between the pressure head and the flow rate of the pump. A standard pump performance map has three curves (Pump Systems Academy, 2025) and an example performance map is displayed in Figure 1:

1. Head vs. Flowrate
2. Power vs. Flowrate
3. Efficiency vs. Flow

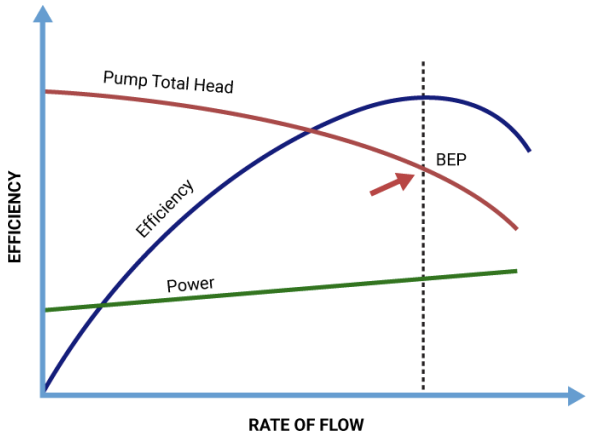


Figure 1: Example of a pump performance map (Pump Systems Academy, 2025).

The point at which the pump's efficiency curve is highest indicates the best efficiency point (BEP), which typically is the basis of the recommended operating range for pumps, as seen in Figure 1. The flow rate at which the peak efficiency is achieved is considered the BEP, and in the end use application, pumps are generally selected by engineers such that their load is within "70 to 120 percent of BEP flow for clear liquids" (Pump Systems Academy, 2025).

The net positive suction head required (NPSHR) is also a key metric that defines the "minimum" pressure required on the inlet of the pump to assure regular operation, as dipping below this range will cause cavitation. On the contrary, the shut off head is a metric that describes the maximum head capacity of the pump, i.e., when the flow rate, Q, is 0. At this point, the pump is at its maximum pressure output and there is no net flow of fluid.

These metrics describe the capabilities of a given pump system and allow engineers to make informed decisions on the best hardware to use for a given application.

Objective

The primary objective of this experiment is to characterize the performance of a sump pump by generating a pump performance curve. This is achieved by experimental determination of the pump's response to various loads by varying voltage input to create a performance map. To achieve this, the following characteristics are measured:

- Flow Rate (L/min), using a pitot tube rotameter-type flow meter
- Pressure (psi), using a pressure gauge immediately downstream of the pump
- Power (W), using a power meter

The height of the main pipe above the reservoir is also measured to account for static head differences.

Materials and Methods

Apparatus

The experimental apparatus consisted of a water reservoir, a sump pump, a regulating valve, a Variac power regulator, a pressure gauge, a flow meter, and a power meter, as seen in Figure 2. The pump was placed within the water reservoir and water was recirculated by the pump through the apparatus back into the reservoir. The regulating valve was used to vary the load conditions of the pump. The sump pump was connected to the power meter, then to the Variac, and the Variac was connected to an AC wall outlet. The Variac power regulator was used to adjust the AC voltage supplied to the pump motor, while the power meter displayed the power consumed in Watts. The flow meter, the pressure gauge, and the power meter were used to monitor and record data for each distinct case.

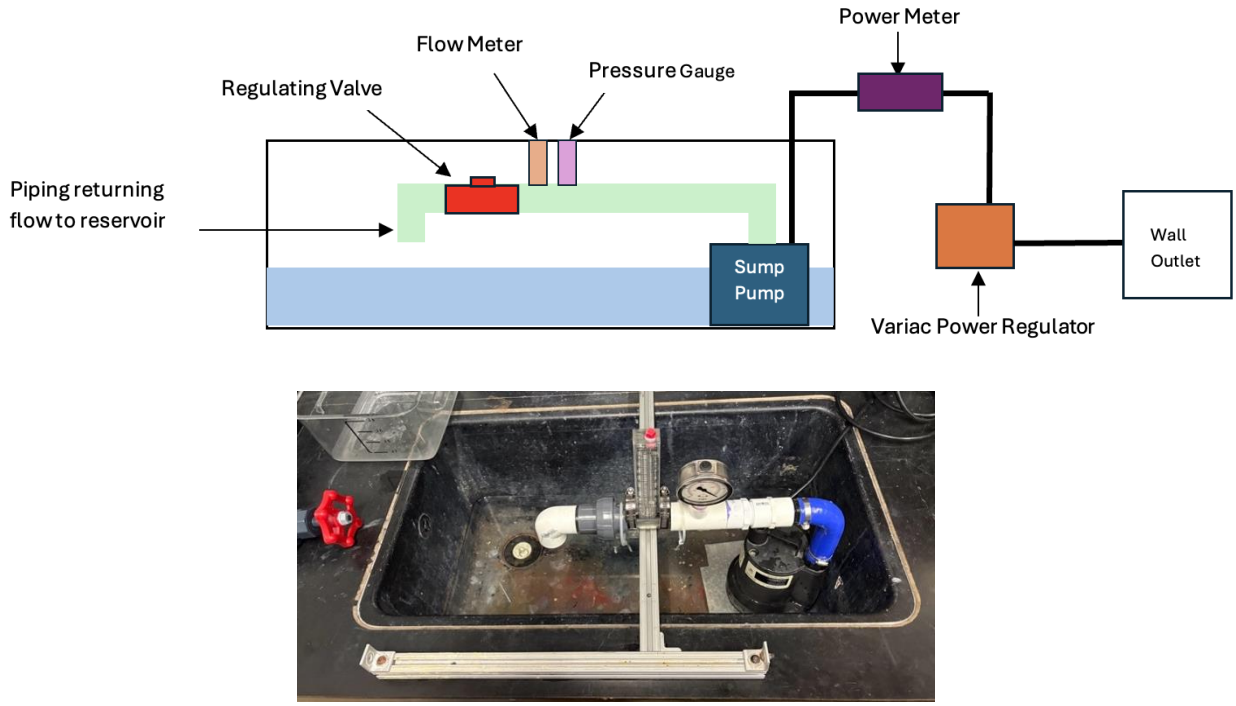


Figure 2: Diagram of Sump Pump Experiment and Picture of Actual Apparatus

The Everbilt 1/6 HP Plastic Utility Pump was used as the test pump, and to measure flow, pressure, and power usage, the F-300 Pilot Tube Flow Meter, NoShok 0-15 psi Pressure Gauge, and “Watts Up” Power Meter were used.

The Everbilt 1/6 HP Plastic Utility Pump is a centrifugal pump that’s listed to work at 115V, has a maximum lift/head of 20 feet, and can pump water that contains solid particles 1/8 inch or less in diameter.

The F-300 Pilot Tube Flow Meter is capable of measuring flow in fluids with a maximum fluid temperature of 190°F / 88°C at 0 psi, a maximum working pressure of 75 psi at 70°F / 21°C, and a measurement range of 2-1900 Gallons per Minute (8-7200 Liter per Minute range) (F-300 Pitot Tube Acrylic Flow Meter, 2026).

The NoShok 0-15 psi Pressure Gauge is capable of primary pressure ranges of 0-15 psi, an ambient and media temperature range of -4-140°F, an accuracy of 1.6% (NoShok Pressure Gauge, 2026).

The "Watts Up" Power Meter is capable of measuring AC voltage from 0 to 120V, a maximum current of 15 amperes, at frequency of 60 Hz, and has an accuracy of $\pm$3% for loads above 10 watts and an accuracy of $\pm$5% for loads below 10 watts (PowerMeterStore, 2026).

Procedure

The test rig was constructed in accordance with the design shown in Figure 2. For the initial measurement, the regulating valve was removed from the system, and the pump outlet was directed into the reservoir. Electrical power to the pump was supplied through a power meter connected to a variable voltage power supply (Variac). The Variac, set to a test voltage, was connected to the wall AC power supply (110V 60Hz). The main pipe was leveled to eliminate hydrostatic pressure differences, and the flow meter was fixed in the vertical position. Before supplying power to the pump, the main pipe centerline height above the reservoir water surface was measured and recorded. The power supply was switched on, and the pressure, flowrate and power were measured and recorded. This data was recorded to characterize the pump performance at minimal resistance (max flow condition).

After this measurement, the regulating valve was reinstalled and fully closed. The pressure, flowrate and power were measured. This step was performed quickly to avoid the pump from overheating. Next, the valve was adjusted to generate intermediate data points for pressure, flow rate, and power. The same procedure was repeated across multiple voltage settings to fully characterize the performance of the pump. The rotameter was calibrated using a volumetric method by collecting water in a container over a measured time interval and calculating the flow rate. Multiple trials were performed and averaged to validate the flow meter readings.

Analysis Procedure

The total dynamic head (TDH) at each operating point is calculated using Equation 1 to generate a TDH versus flow rate curve where the pressure head is defined in terms of the gauge pressure p, the density of water ρ =0.9982 $\frac{g}{cm^3}$ at 20°C $1atm$, gravitational acceleration g = $\frac{9.81m}{s^2}$, the difference z_{cl} between the main pipe centerline and the reservoir water surface, friction head H_f and velocity head H_v. However, as inlet and outlet pipe diameters are constant and velocities are low, and when minimal major and minor losses are assumed to be negligible, $H_v \approx H_f \approx 0\ m$.

$$H_p = H_p + Z_{cl} + H_f + H_v \quad (1)$$

Electrical power was also analyzed as a function of flow rate to identify trends in motor demand to identify efficient and inefficient regions. Pump efficiency, η, was calculated as the ratio of the hydraulic power delivered to the fluid, given by the product of the pressure rise across the pump $\Delta\, p$ and the volumetric flow rate Q, to the electrical input power to the pump motor P_{elec} (Equation 2).

$$\eta = \frac{\rho g Q h_p}{P_{elec}} \quad (2)$$

Experimental Data

Data Summary

Data was collected across multiple experimental teams. Each team was assigned a specific voltage to be used as the voltage supplied to the pump; the teams and the assigned voltages are displayed in Table 1. Teams replicated the exact experimental procedure for the assigned test voltage and obtained data points for pressure, flow, and power consumed (from max flow, zero flow, and multiple intermediate data points).

The data was aggregated and analyzed to obtain the desired pump performance map for each voltage setting. The varying voltages are tested to find the optimal point for efficiency based on voltage and to simulate conditions in which the pump experiences voltage drops. From data obtained for each group, the range of flow rates was from 0 LPM to 86.5 LPM, the range of pressures was from 1 psi to 10.5 psi, and the range of power draw was from 146 W to 305 W.

Table 1: Summary of Cases Run by Each Team

Team	Variac Setting (V)
A1	125
A2	110
A3	95
B4	100
B5	115
B6	120
A7	120

Pressure vs Flow

Figure 3 shows the raw pressure versus flow rate data for all trials. Across all voltage settings, pressure decreases as flow rate increases, demonstrating the expected inverse relationship for pump performance. Additionally, the pressure–flow curves are nearly identical across different voltage levels, indicating that voltage has minimal effect on pump performance within the tested range. The plots with error bars show that measurement uncertainty is relatively small, and the consistency of the data across trials suggests good repeatability of the experimental measurements.

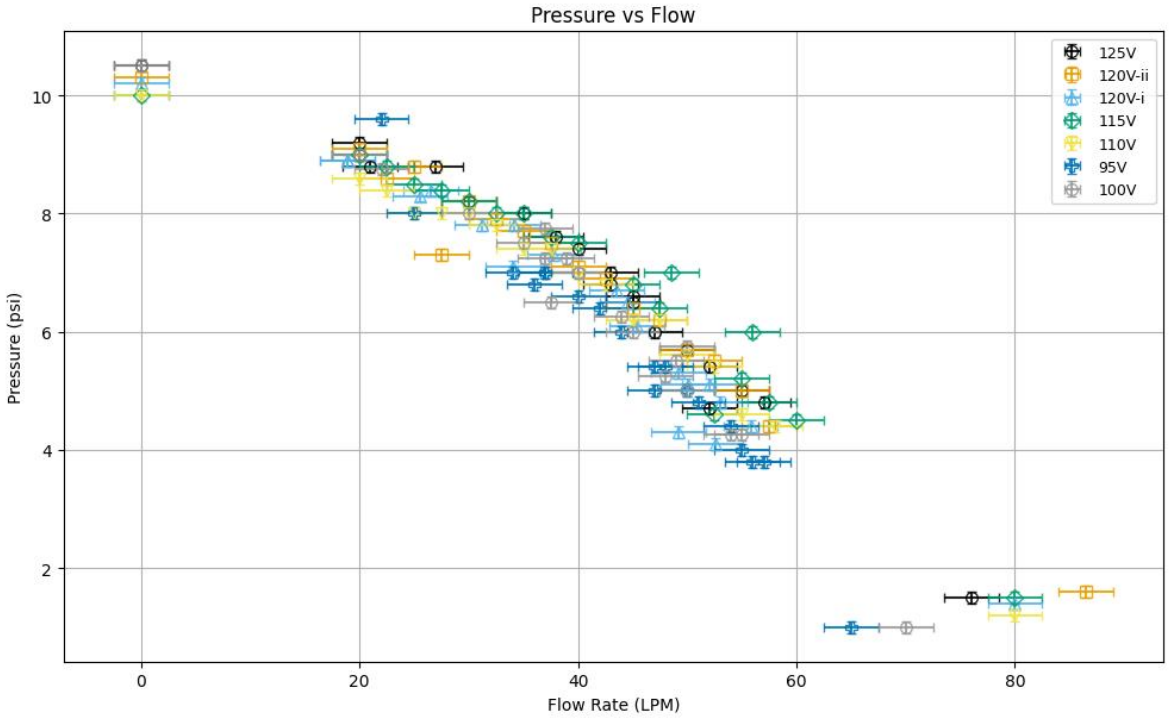


Figure 3: Raw Data of all Tests for Pressure as a Function of Flow Rate

Figure 4 presents the pressure–flow data along with quadratic lines of best fit for each case. The curves closely follow the experimental data, indicating that a quadratic model adequately represents the pump characteristic curve. High R^2 values show strong agreement between the model and the measured data, confirming that the pressure decreases nonlinearly as flow rate increases for each voltage condition.

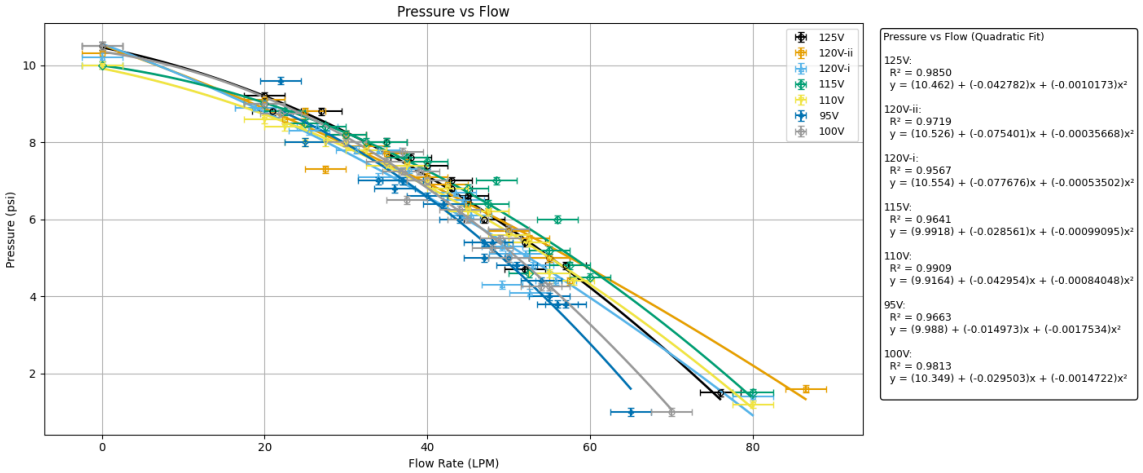


Figure 4: Raw Data for Pressure versus Flow Rate with Quadratic Line of Best Fit

Power vs Flow

Figure 5 shows raw electrical power consumption as a function of flow rate for all trials. In general, power increases with increasing flow rate across all Variac settings. The error bars indicate that the measurement uncertainty is relatively small compared to the overall trend, and the separation between voltage levels shows that higher supply voltages result in higher power consumption throughout the operating range.

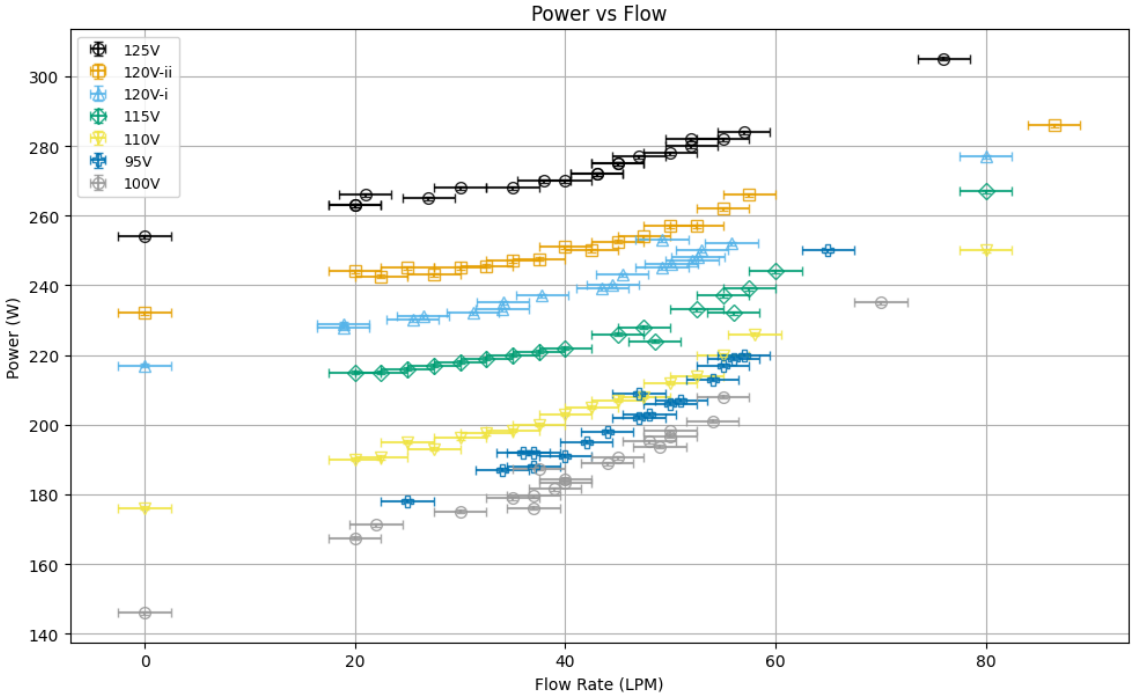


Figure 5: Raw Data of all Tests for Power as a Function of Flow Rate

Figure 6 represents power–flow data with quadratic lines of best fit for each voltage condition. The curves closely follow the experimental data, indicating that the quadratic model adequately represents the relationship between power consumption and flow rate. The high R^2 values suggest strong agreement between the model and the measurements, confirming that pump power demand increases nonlinearly with increasing flow rate. Outlier points identified during testing were removed prior to analysis following discussion with the advisor to ensure a more accurate representation of the system behavior.

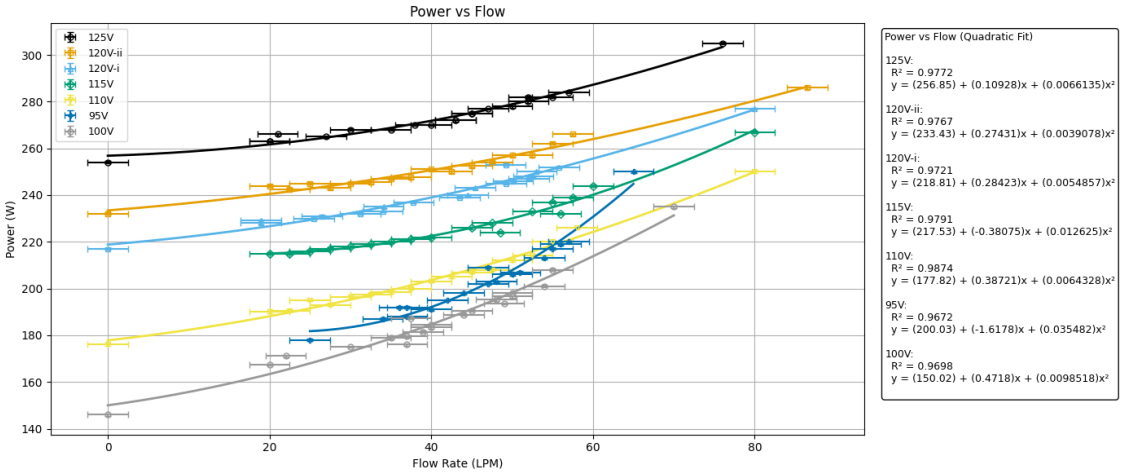


Figure 6: Raw Data for Power versus Flow Rate with Quadratic Line of Best Fit

Sources of Error

The dominant uncertainties in this experiment arise from instrument resolution and short-term flow variability. The flow meter (rotameter) is graduated in 5 LPM increments, giving 2.5 LPM measurement uncertainty. Particularly, it was noted that the flow rate was variable to 2 LPM due to turbulence within the pipe and that there was a full-scale accuracy of $\pm 5\%$ for pipes 2” in diameter or less (F-300 Pitot Tube Acrylic Flow Meter, 2026). While a median value was selected for recordkeeping, this further contributes to variations in the results. The pressure gauge was equipped with 0.2 psi graduations, as such ± 0.10 psi of measurement uncertainty was assigned. While minimal, the manufacturer also lists an accuracy of $\pm 1.6\%$ which also impacts the measurements (NoShok Pressure Gauge, 2026). The pipe centerline height, measured with a tape marked to 1/16 inch, contributes plus or minus 1/32 inch of measurement uncertainty, which corresponds to only approximately 0.0011 psi of hydrostatic head in water and is thus negligible relative to the gauge resolution. The power meter used also had its own uncertainty of ± 1 W and with an accuracy of $\pm 3\%$ for loads over 10 W (PowerMeterStore, 2026). Additional factors include parallax when reading analog dials and slight zeroing/alignment errors when applying the centerline offset.

From the repeated bucket and stopwatch calibrations (Table 2), the deviation of the measured flow is typically approximately 3.3 LPM (1 sigma) across representative five trial sets at the 40 to 60 LPM setpoints, capturing timing error and short-term flow fluctuations. Combining this repeatability (plus or minus 3.3 LPM) with rotameter uncertainty (plus or minus 2.5 LPM) in quadrature yields a representative overall flow uncertainty of about plus or minus 4.1 LPM. For pressure, the gauge resolution (plus or minus 0.10 psi) dominates the budget, with the head measurement contribution (approximately plus or minus 0.001 psi) effectively negligible. Small non monotonic variations in neighboring flow/pressure points—consistent with pump and power supply variability—but the effect is encompassed by the measured flow repeatability above.

Table 2: Summary Flow Calibration

	58LPM		40 LPM		50 LPM		21 LPM		30 LPM		76 LPM	
trial #	Volume (L)	Time (s)	Volume (L)	Time (s)	Volume (L)	Time (s)	Volume (L)	Time (s)	Volume (L)	Time (s)	Volume (L)	Time (s)
1	4	3.2	4	6.06	3	3.79	1.9	5.02	3	5.48	2.67	2.03
2	4	3.86	3	5.33	3	3.57	1.7	4.69	4	7.39	3.5	2.43
3	4	3.48	3	5.23	3	3.67	3.2	8.24	2	3.48	2.5	1.93
4	4	3.26	2	2.91	3	3.34	2.6	6.74	3	5.26	3.5	2.4
5	4	3.83	1	1.71	3	3.57	3.1	8.03	4	7.02	2.67	2.01

Data Analysis

Pump Curves

Head vs Flow

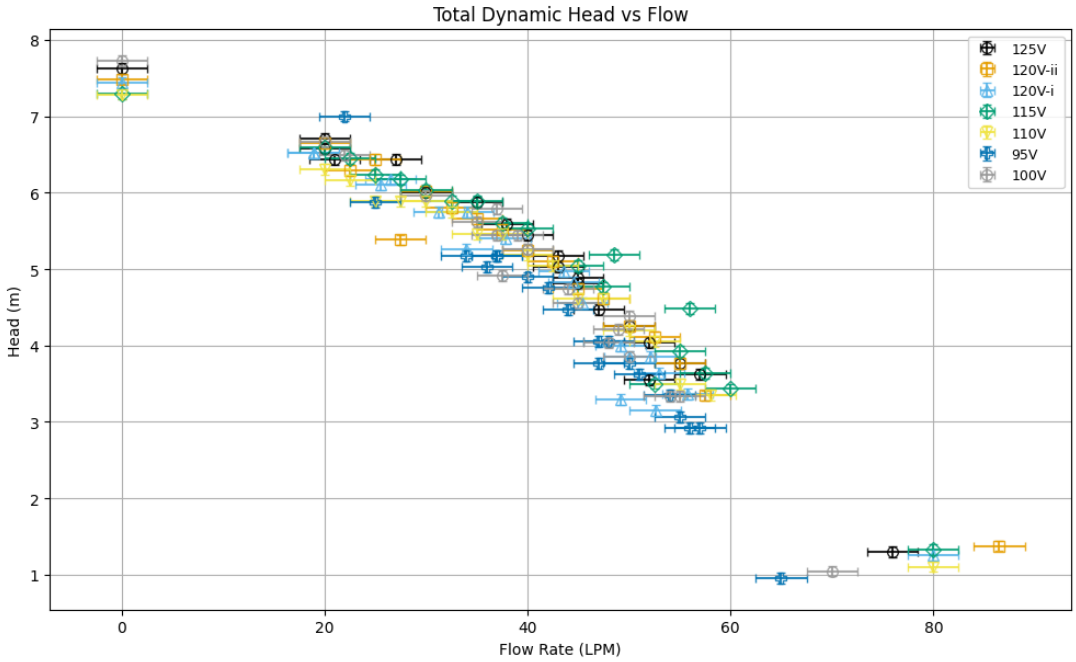


Figure 7: Raw Data of all Tests for Total Dynamic Head as a Function of Flow Rate

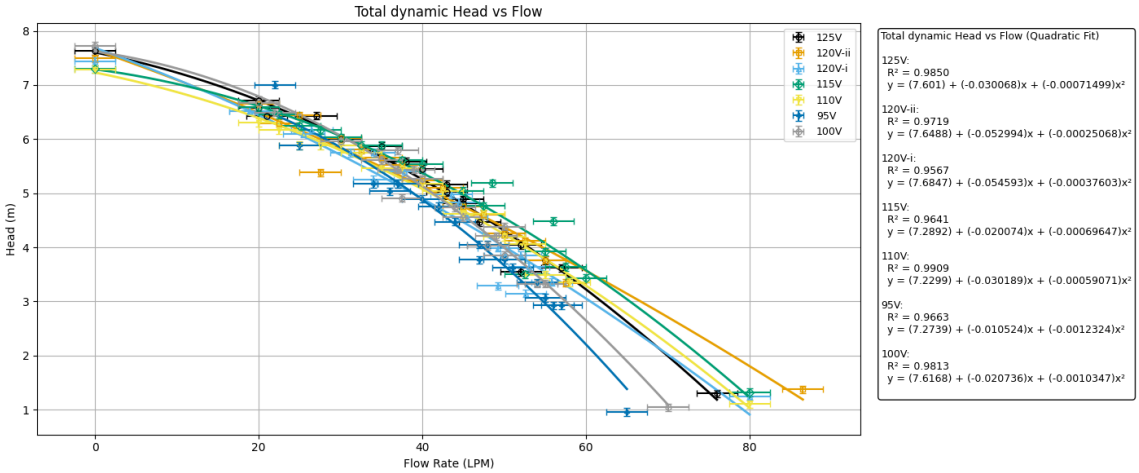


Figure 8: Raw Data for Total Dynamic Head vs Flow Rate with Quadratic Line of Best Fit

The regression analysis of the pump performance data shows that the polynomial models provide a strong fit for both the total dynamic head and efficiency relationships. The cubic models frequently produce higher R-squared values, often exceeding 0.98, but their Cross-Validation R-squared values are lower. Cross-Validation R-squared is a metric that evaluates how well a model predicts unseen data by testing its performance on

validation subsets that are excluded during model fitting through repeated train–test splits (James, Witten, Hastie, & Tibshirani, 2021). This distinction is important as it reflects the model's ability to generalize beyond the measured data and helps detect overfitting. The lower Cross-Validation R-squared for the cubic models indicates that they are likely to overfit the data rather than accurately represent the physical behavior of the pump. In contrast, the quadratic models maintain high standard R-squared values while also demonstrating strong Cross-Validation R-squared scores, confirming that they provide a more reliable and generalizable representation of pump performance across different voltage test conditions.

To verify the physical validity of the quadratic model, fluid dynamic modeling of the pump is required. Fundamentally, for an ideal Eulerian centrifugal pump where the fluid is frictionless and the flow is perfectly uniform, the head generated by the pump is given by the equation:

$$H_{pump} = \frac{U_{out}V_{out} - U_{in}V_{in}}{g} \quad (3)$$

where U is the tangential speed of the impeller blade, and V the tangential speed of the fluid. In most systems and in the testing apparatus, the fluid enters the pump straight, meaning $V_{in} = 0$:

$$H_{pump} = \frac{U_{out}V_{out}}{g}$$

To compute flow rate from this result, we consider the velocity at the exit, where the fluid is moving outward radially but also dragged tangentially by the impeller:

- U_{out}: speed of impeller blade tip
- $V_{r_{out}}$: radial velocity of fluid
- β_{out}: angle of impeller blade at exit

We define the tangential fluid velocity from (3) as the blade tip speed minus the slip from the fluid in contact with the back of the blade:

$$V_{out} = U_2 - V_{r_{out}} \cot(\beta_{out}) \quad (4)$$

In a standard pump, the outlet is a pipe with a cylindrical surface. Thus, for an impeller with a diameter D and width b, the exit area is computed as $A = \pi Db$. Therefore, the radial velocity is the flow rate divided by the area:

$$V_{r_{out}} = \frac{Q}{\pi Db} \quad (5)$$

and by substituting (5) into (4) and (3):

$$H_{pump} = \frac{U_{out}^2}{g} - \left[\frac{U_{out}\cot(\beta_{out})}{g\pi Db}\right]Q \ (6)$$

as these are given as fixed physical constants, Euler's pump equation can be generalized to the form:

$$H = A - BQ \ (7)$$

However, this is for an ideal pump with no friction losses. Friction head losses in pipe systems are given by the Darcy-Weisbach equation:

$$h_f = f\frac{L}{D}\frac{v^2}{2g} \ (8)$$

given that $v = \frac{Q}{A}$ where Q is the volumetric flow rate and A is the cross-sectional area of the pipe, we obtain:

$$h_f = CQ^2 \ (9)$$

where C represents a constant given by the physical parameters of the system. Therefore, the overall system head is given as:

$$H_{sys} = H_{static} + CQ^2 \ (10)$$

Combined with the generalized constants A and B in (7), the final model for the head produced by pump is given by a model described as a quadratic formula:

$$\boldsymbol{H = A - BQ - CQ^2} \ (11)$$

Thus, a quadratic model is the most physically sound regression model to characterize the pump head against flow rate.

Efficiency vs Flow

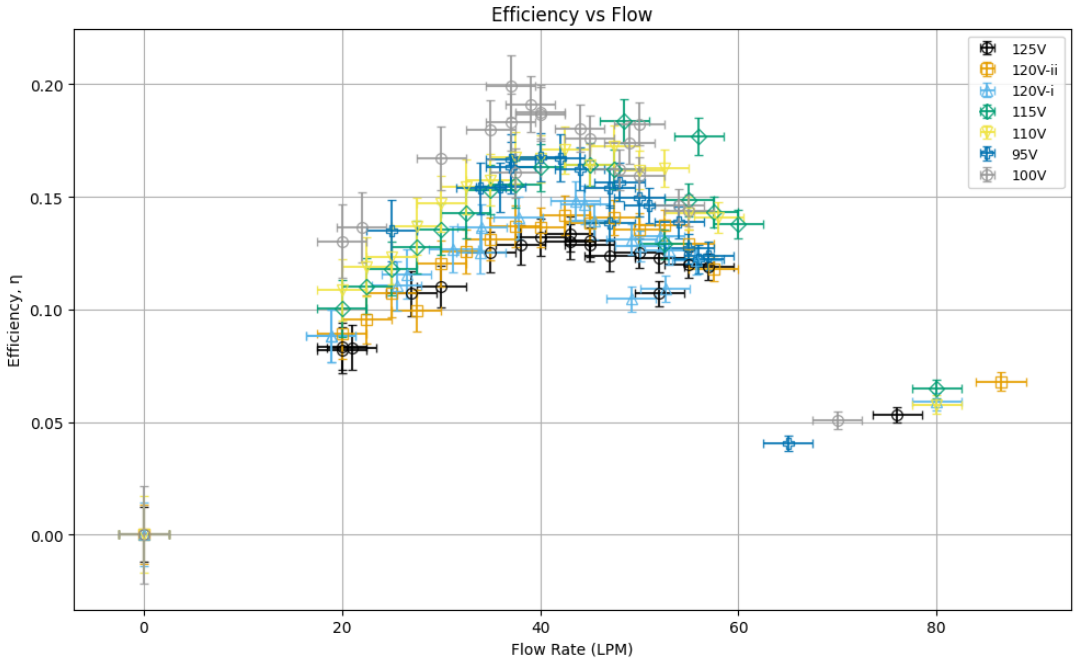


Figure 9: Raw Data for Efficiency as a Function of Flow Rate

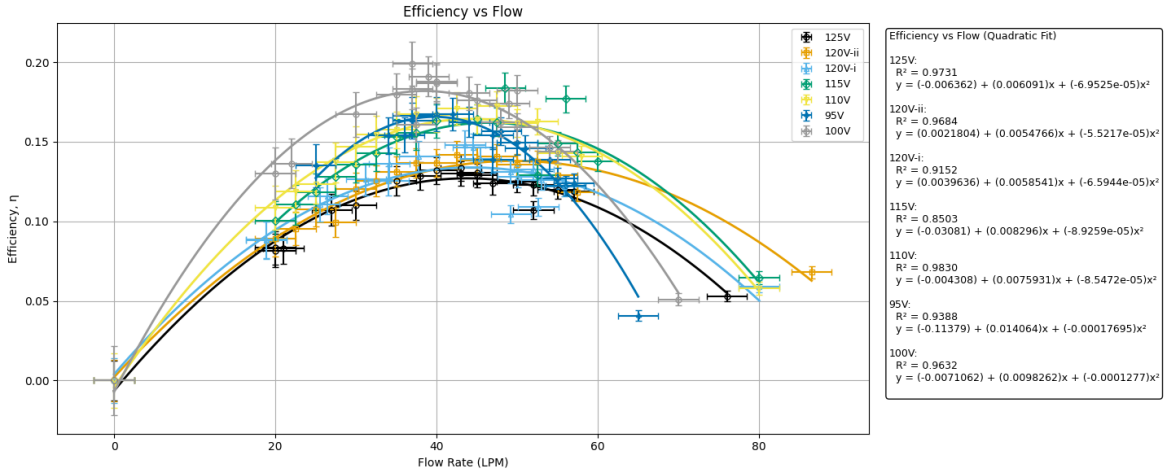


Figure 10: Raw Data for Efficiency versus Flow Rate with Quadratic Line of Best Fit

The efficiency versus flow rate data typically follows a parabolic shape. The quadratic regression models represent this behavior, allowing for the calculation of the Best Efficiency Point. The negative quadratic coefficients in all the efficiency equations confirm the shape of the curve, representing the physical condition that efficiency decreases when the pump operates outside the optimal flow range.

The standard R-squared values for the efficiency quadratic models are generally high, often above 0.90, but the Cross-Validation R-squared scores show more variation than

the scores for the head curves. This variation indicates that efficiency calculations are highly sensitive to measurement uncertainties in power consumption, pressure, and flow rate. The quadratic models remain the most appropriate choice because linear models fail to calculate the maximum efficiency point, and cubic models introduce unrealistic directional changes in the curve.

Operating Conditions

The best efficiency point (BEP) for this pump is determined by calculating the local maximum of the quadratic efficiency regressions. By determining flow rates where the derivative of the efficiency curve equals zero, the optimal operating zone (BEP) is identified. Across the test voltages from 110V to 125V, the BEP consistently is between 35 to 50 liters per minute. While the 100V test case offers the highest nominal efficiency metric of all trials at 18.19%, the flow rate at this point is notably lower at 38.47 LPM. Due to the overall low efficiency of the pump across all testing, a marginally higher efficiency metric may not be practically useful; a better metric would account for both the flow and efficiency.

Thus, to better evaluate the results, a specific energy consumption (SEC) quotient is calculated for each BEP to compare the flow rate relative to efficiency, which we define as the unit amount of power required to move a specific volume of fluid:

$$SEC = \frac{P_{in}}{Q}$$

Using the Power vs. Flow Rate models, we can evaluate the test cases at each BEP to compute the expected power draw at the BEP. This computation yielded Table 2:

Table 3: Evaluation Metric Comparison

Voltage	BEP Flow Rate (LPM)	BEP Efficiency (%)	BEP Power Draw (W)	SEC (W/LPM) (lower is better)	Head (m)
100V	38.47 ± 0.62	18.19 ± 0.003	182.8 ± 7.4	4.75 ± 0.21	5.288 ± 0.471
110V	44.42 ± 0.52	16.43 ± 0.002	207.7 ± 4.5	4.68 ± 0.11	4.723 ± 0.328
115V	46.47 ± 1.03	16.2 ± 0.004	227.1 ± 8.8	4.89 ± 0.22	4.852 ± 0.700
120V(i)	44.39 ± 1.16	13.39 ± 0.003	242.2 ± 5.0	5.46 ± 0.18	4.520 ± 0.733

In conclusion, it is evident that the 100V operating voltage gives the best-case result for this pump. While operating at 100V, the sump pump requires 4.75W of energy to move 1 LPM of fluid. Among all trials, the 110V and 100V case both offer the best nominal SEC scores at the BEP; however, the uncertainty reveals that these values are statistically indistinguishable.

To determine the best operating mode, the efficiency at the BEP is considered, in which the 100V case presents a statistically significant best in class efficiency. While a push to higher voltages can yield marginally higher flow rates, for the expected use case of a sump pump, higher energy savings and safer operating limits for increased reliability will be more desirable than a marginally quicker pumping cycle.

Thus, it is recommended that this pump be used at 100V, in applications where the flow requirements lie between 35 to 45 LPM. Operating the pump outside this range will reduce efficiency and introduce likelihood of premature failure of the pump and increase utility costs.

Conclusion

Summary

Seven groups ran experiments on the AC pump apparatus for voltages from 95V-125V, each obtaining data for flow rate, pressure, and power consumed. After these data points were collected, it was used to calculate head versus flow rate and efficiency versus flow rate. After analysis, it was determined that head versus flow rate should best be represented by a quadratic line of best fit. The head versus flow rate plots having a quadratic line of best fit indicate predictive reliability on the values of head across multiple voltage cases, being in line with the theory that head would decrease as flow rate increased. When efficiency and flow rate were analyzed, it was also determined that a quadratic fit would be suitable, meaning that efficiency was given as a nonlinear function of flow. The system was highly sensitive to measurement uncertainties of the three measurements, indicated by large variation in CVR^2 values. When the best efficiency point (BEP) was calculated, the ideal operation regime was identified to be at 38-47 LPM; however, to better account for differences in flow rates at the BEP, the specific energy consumption quotient was computed. After comparative analysis of all scoring factors for best efficiency and functional throughput, it was determined that the 100V mode presented the best results.

Recommendations

It can be said that the primary sources of instability were measurement errors and system instability. To minimize uncertainties, better methods of control and better instrumentation are recommended. By using a higher resolution flow meter, the uncertainty can be reduced from the currently large $\pm$ 5 LPM uncertainty. Digital pressure sensors could also provide better resolution and allow for automatic logging of data, minimizing uncertainty and human error. Particularly, continuous measurement can allow for normalization of the fluctuations in pressure. Finally, using a fine control valve would allow for more precise and repeatable flow adjustment to allow for higher resolution data across flow rates. Finally, allowing the system to reach steady state before data collection can ensure that the data is as accurate as possible.

Data Quality

The trends in data in this experiment indicate that the metrics were relatively precise in most test cases. However, the 95V test case had significantly deviant data, considered erroneous outlier values that were removed for the purposes of this experiment. When comparing the results to the published values, the published values

provided a larger flow rate for the head value than what was calculated and is not encompassed by the uncertainty of the calculations. Manufacturer test data indicates pump testing at 115V and 2.9A, yielding 1212 gallons per hour (76.47 LPM) with 10ft (3.048m) of head (Everbilt). When compared to the flow rate and head value of the 115V case, at 3.43m $\pm$ 0.0703 m head, flow rate is approximately 60 LPM $\pm$ 2.5 LPM. While this was the most proximal data point to manufacturer data, there is significant deviation between the data from the experiment even after accounting for error.

This suggests that the results of the experiment are relatively inaccurate compared to published values; however, these differences from the experimental and the published data could be due to the difference in testing conditions, the equipment used, the setup of the testing system, and accuracy of the sensors. The very worn-down condition of the unit used in the experiment could suggest a strong reason for lower flow rate relative to published specifications.

Appendix

Appendix 1: Raw Data

120 V - i			
#	Flow Rate (LPM)	Pressure (psi)	Power (Watts)
1	0	10.2	217
2	18.9	8.9	228
3	18.9	8.9	229
4	25.5	8.3	230
5	26.5	8.4	231
6	31.23	7.8	232
7	34	7.1	233
8	34.1	7.8	235
9	37.8	7.3	237
10	43.53	6.7	239
11	44.5	6.5	240
12	45.42	6.1	243
13	49.21	5.3	245
14	50.1	5.1	246
15	52.05	5.1	247
16	52.6	4.1	248
17	53	4.8	250
18	55.8	4.4	252
19	49.2	4.3	253
20	80	1.4	277

120 V - ii			
#	Flow Rate (LPM)	Pressure (psi)	Power (Watts)
1	86.5	1.6	286
2	0	10.3	232
3	20	9.1	244
4	25	8.8	245
5	30	8.2	245
6	35	7.7	247
7	40	7.1	251
8	45	6.4	252.5
9	50	5.7	257
10	55	5.0	262
11	57.5	4.4	266
12	52.5	5.5	257
13	47.5	6.2	254
14	42.5	6.9	250
15	37.5	7.5	247.5
16	32.5	7.9	245.5
17	27.5	7.3	243
18	22.5	8.6	242.5

115 V			
#	Flow Rate (LPM)	Pressure (psi)	Power (Watts)
1	80	1.5	267
2	57.5	4.8	239
3	52.5	4.6	233
4	60	4.5	244
5	55	5.2	237
6	56	6	232
7	45	6.8	226
8	40	7.5	222
9	35	8	220
10	30	8.2	218
11	25	8.5	216
12	20	9	215
13	22.5	8.8	215
14	27.5	8.4	217
15	32.5	8	219
16	37.5	7.6	221
17	48.5	7	224
18	47.5	6.4	228
19	0	10	264

110 V			
#	Flow Rate (LPM)	Pressure (psi)	Power (Watts)
1	80	1.2	250
2	0	10	176
3	30	8	196.2
4	40	7	203
5	50	5.6	212
6	55	4.6	220
7	58	4.4	226
8	45	6.2	207
9	35	7.4	198.4
10	25	8	195
11	20	8.6	190
12	22.5	8.4	190.6
13	27.5	8	193
14	32.5	7.8	197.5
15	37.5	7.4	200
16	42.5	6.8	205
17	47.5	6.2	208
18	52.5	5.4	214

95 V			
#	Flow Rate (LPM)	Pressure (psi)	Power (Watts)
1	65	1	250
2	26	10	150
3	47	5	209
4	37	7	192
5	22	9.6	150
6	36	6.8	192
7	42	6.4	195
8	54	4.4	213
9	56	3.8	219
10	57	3.8	220
11	51	4.8	207
12	40	6.6	191
13	50	5.0	206
14	55	4.0	217
15	25	8.0	178
16	34	7.0	187
17	37	7.0	188
18	48	5.4	203
19	44	6.0	198
20	47	5.4	202

100 V			
#	Flow Rate (LPM)	Pressure (psi)	Power (Watts)
1	70	1	235
2	0	10.5	146
3	20	9	167.4
4	30	8	175.1
5	35	7.5	178.8
6	40	7	184.4
7	45	6	190.5
8	50	5	198.1
9	55	4.25	208
10	48	5.25	195.2
11	44	6.25	188.8
12	40	7	183.3
13	37	7.25	179.6
14	22	8.75	171.2
15	37	7.75	176
16	37.5	6.5	187.4
17	50	5.75	196.6
18	54	4.25	201
19	49	5.5	193.6
20	39	7.25	181.5

125 V			
#	Flow Rate (LPM)	Pressure (psi)	Power (Watts)
1	76	1.5	305
2	0	10.5	254
3	27	8.8	265
4	35	8	268
5	45	6.6	275
6	47	6	277
7	52	5.4	280
8	55	5	282
9	57	4.8	284
10	20	9.0	263
11	38	7.6	270
12	43	7	272
13	20	9.2	263
14	21	8.8	266
15	30	8.2	268
16	40	7.4	270
17	45	6.5	275
18	50	5.7	278
19	43	6.8	272
20	52	4.7	282

Appendix 2: Instrument Full Specs

No Shok Pressure Gauge – Key Features	
Accuracy	±1.6%
Primary Pressure Range	0-15 psi
Ambient Temperature Range	-4-140°F
Media Temperature Range	-4-140°F
Process Connection Size	¼ in. NPT or 7/16 in.-20 Adjustable

F-300 Pitot Tube Acrylic Flow Meter – Key Features	
Accuracy	±5% for L2 Pipe Size
Max Working Pressure	75 psi @ 70°F
Max Fluid Temperature	190°F @ 0 psi
Flow Rate Range	8-7200 LPM

"Watts Up Pro" Power Meter – Key Features	
Accuracy	±3% (loads above 10W) ±5% (loads below 10W)
Power Requirements	120V AC, 60 Hz, 15 amps
Mains Supply Voltage	Fluctuations not to exceed ±10% of nominal voltage
Usage	Indoor Usage Only

Everbilt 1/6 HP Plastic Utility Pump – Key Features	
Nominal Operating Voltage	115 V
Max Head	20ft
Compatible Media	Water
Accuracy	Not Provided by Manufacturer
Operating Current	2.9A
Horsepower	1/6

Appendix 3: Sample Dynamic Head and Efficiency Calculation

Using one 100V datapoint

Flow Q = 40 LPM

Gauge Pressure P = 7 psi

Power $P_{elec} = 203W$

Centerline height $Z_{cl} = 10.25\ in = 0.26m$

Since inlet, outlet pipe diameters are similar and velocities aren't very high, $v = 0m/s$.

Friction head is assumed to be negligible, $H_f = 0$.

Converting pressure psi to Pa

$$\Delta p = 7\ psi\ \times \frac{6894.76\ Pa}{1\ psi} = 48263.32\ Pa$$

Computing dynamic head

$$H_p = \frac{\Delta p}{\rho g} + z_{cl} + H_f + H_v$$

$$H_p = \frac{\Delta p}{\rho g} + z_{cl} + 0 + \frac{v^2}{2g} = \frac{48263.32\ Pa}{\left(\frac{1000kg}{m^3}\right)\left(\frac{9.81m}{s^2}\right)} + 0.26m + 0 + 0 = 5.18m$$

Converting flow LPM to $\frac{m^3}{s}$

$$\text{Q} = 40\text{LPM}\ \times \frac{1\text{m}}{60\text{s}} \times \frac{1m^3}{1000\ L} = \frac{0.0006667m^3}{s}$$

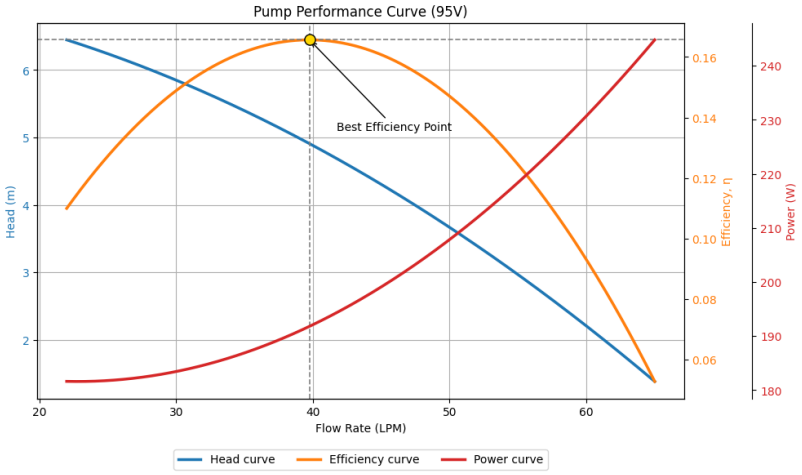
Computing hydraulic power

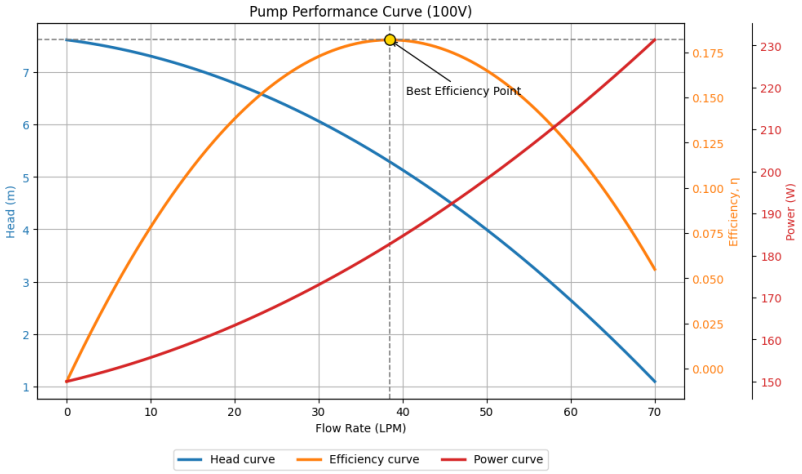
$$P_{hyd} = \rho g Q H_p = \left(\frac{1000kg}{m^3}\right)\left(\frac{9.81m}{s^2}\right)\left(\frac{0.0006667m^3}{s}\right)(5.18m) = 33.9W$$

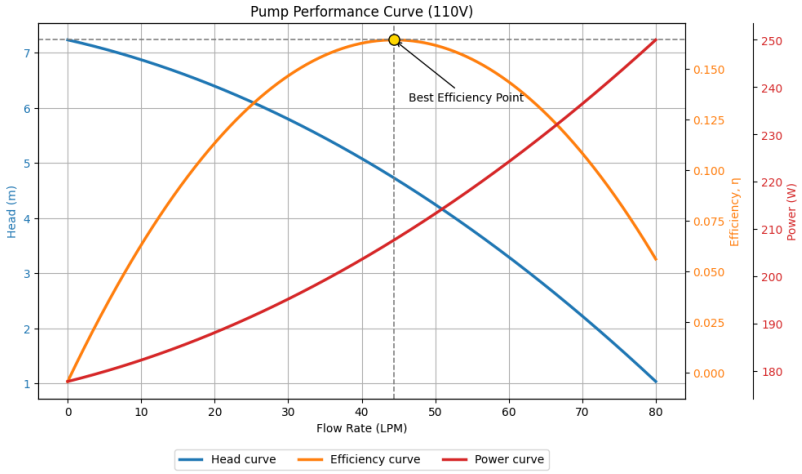
Computing Efficiency

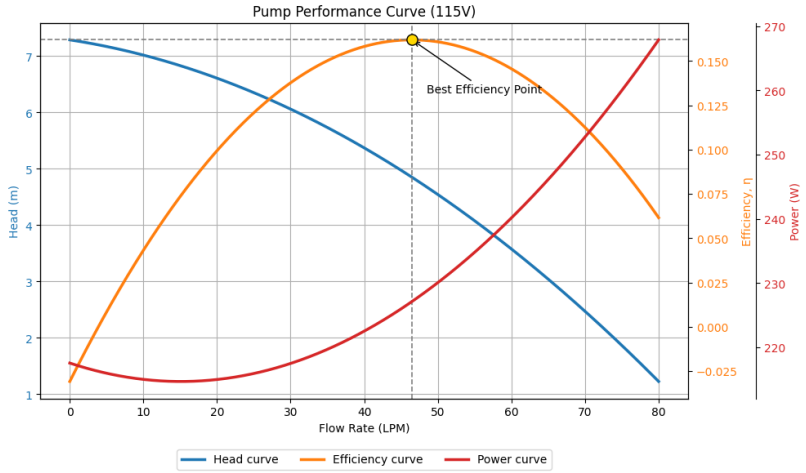
$$\eta = \frac{P_{hyd}}{P_{elec}} = \frac{33.9W}{203W} = 16.7\%$$

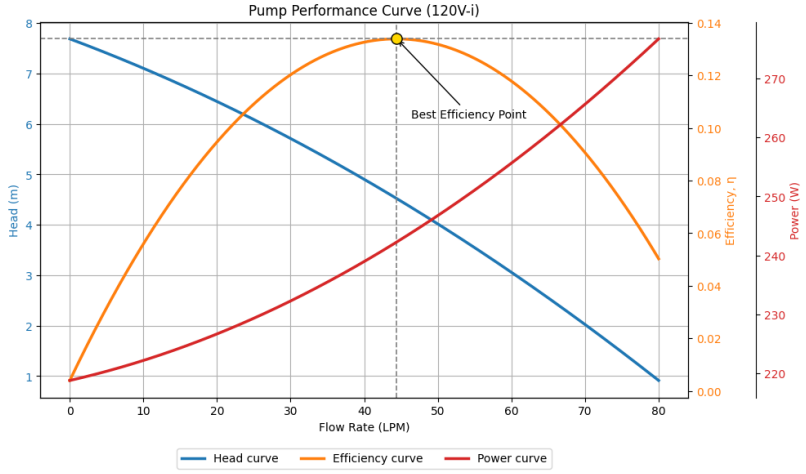
Appendix 4: Pump Performance Curves to Find Specific Energy Consumption (SEC)

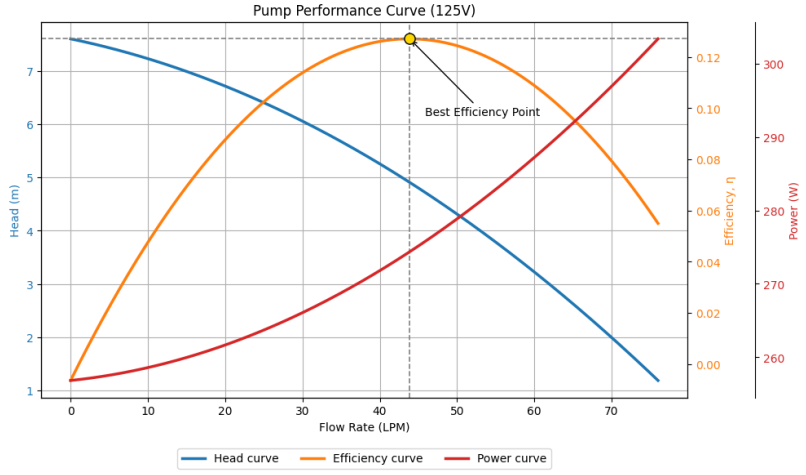







Pump Performance Curve (115V)
Best Efficiency Point
Head (m)
Efficiency, η
Power (W)
Flow Rate (LPM)
Head curve
Efficiency curve
Power curve


Pump Performance Curve (120V-i)
Best Efficiency Point
Head (m)
Efficiency, η
Power (W)
Flow Rate (LPM)
Head curve
Efficiency curve
Power curve


Pump Performance Curve (125V)
Best Efficiency Point
Head (m)
Efficiency, η
Power (W)
Flow Rate (LPM)
Head curve
Efficiency curve
Power curve

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